Week 11–12 — Risk Measures: Mean–Variance, VaR, CVaR

AD7031 Optimization Under Uncertainty • Instructor: Prof. H. Park • Nov 7 & 14, 2026

Recap: chance constraints and the Mixed Integer Program (MIP) (week 10)

Chance-constrained program $\tilde{\xi} \sim \mathbb{P}$ is a random vector, $\mathcal{X}$ collects the constraints with no uncertainty

$$\begin{aligned} \min_{x \in \mathcal{X}} \quad & f_0(x) \\ \text{s.t.} \quad & \mathbb{P}\big(f_i(x, \tilde{\xi}) \le 0\big) \ge 1 - \varepsilon \quad \forall i \in [I] \end{aligned}$$

each constraint must hold with probability at least $1 - \varepsilon$

MIP reformulation

the same steps as the 0–1 loss in week 10: first write it as a cap on the probability of *violation* ($f_i > 0$ is the complement of $f_i \le 0$):

$$\mathbb{P}\big(f_i(x, \tilde{\xi}) \le 0\big) \ge 1 - \varepsilon \quad \Longleftrightarrow \quad \mathbb{P}\big(f_i(x, \tilde{\xi}) > 0\big) \le \varepsilon$$

assume $\mathbb{P}$ is discrete: $\tilde{\xi}$ takes S values ξ^s with probabilities p_s (data: $p_s = \frac{1}{S}$, as in week 10)
then the violation probability adds up the scenarios that violate the constraint:

$$\mathbb{I}[f_i(x, \tilde{\xi}) > 0] = \begin{cases} 1, & \text{if } f_i(x, \tilde{\xi}) > 0 \quad \text{(i.e., } f_i \le 0 \text{ is violated)} \\ 0, & \text{otherwise} \end{cases}$$

$$\mathbb{P}\big(f_i(x, \tilde{\xi}) > 0\big) \le \varepsilon \quad \Longleftrightarrow \quad \sum_{s \in [S]} p_s \, \mathbb{I}[f_i(x, \xi^s) > 0] \le \varepsilon$$

step 1 — one binary per scenario, $z_{is} \ge \mathbb{I}[f_i(x, \xi^s) > 0]$, $z_{is} \in \{0, 1\}$: $\sum_{s \in [S]} p_s \, z_{is} \le \varepsilon$

step 2 — Big-M: $z_{is} \ge \mathbb{I}[f_i(x, \xi^s) > 0]$ becomes $f_i(x, \xi^s) \le M z_{is}$, with $M \ge \max_{x \in \mathcal{X}} f_i(x, \xi^s)$

$z_{is} = 0 \;\Rightarrow\; f_i(x, \xi^s) \le 0$ scenario s must satisfy the constraint
$z_{is} = 1 \;\Rightarrow\; f_i(x, \xi^s) \le M$ always true: scenario s may violate it

the MIP

$$\begin{aligned} \min \quad & f_0(x) \\ \text{s.t.} \quad & f_i(x, \xi^s) \le M z_{is} && \forall i \in [I],\ s \in [S] \\ & \sum_{s \in [S]} p_s \, z_{is} \le \varepsilon && \forall i \in [I] \\ & z_{is} \in \{0, 1\} && \forall i \in [I],\ s \in [S] \\ & x \in \mathcal{X} \end{aligned}$$

Direct use of the chance constraint

$$\mathbb{I}[f_i(x, \tilde{\xi}) \le 0] = \begin{cases} 1, & \text{if } f_i(x, \tilde{\xi}) \le 0 \\ 0, & \text{otherwise} \end{cases}$$

$$\begin{aligned} & \mathbb{P}\big(f_i(x, \tilde{\xi}) \le 0\big) \ge 1 - \varepsilon \\ \Longleftrightarrow \quad & \sum_{s \in [S]} p_s \, \mathbb{I}[f_i(x, \xi^s) \le 0] \ge 1 - \varepsilon \end{aligned}$$

$$\begin{aligned} \min \quad & f_0(x) \\ \text{s.t.} \quad & f_i(x, \xi^s) \le M(1 - z_{is}) && \forall i, s \\ & \sum_{s \in [S]} p_s \, z_{is} \ge 1 - \varepsilon && \forall i \\ & z_{is} \in \{0, 1\} && \forall i, s \\ & x \in \mathcal{X} \end{aligned}$$

here $z_{is} = 1$ means scenario s must satisfy the constraint

1. From deterministic to stochastic optimization

Deterministic program (ξ known)

$$\begin{aligned} \min \quad & f_0(x,\xi) \\ \text{s.t.} \quad & x \in \mathcal{X} \\ & f_i(x,\xi) \le 0 \qquad \forall i \in [I] \end{aligned}$$

Stochastic program ($\tilde{\xi} \sim \mathbb{P}$)

$$\boxed{\begin{aligned} \min \quad & \mathcal{R}_{\mathbb{P}}\big[f_0(x,\tilde{\xi})\big] \\ \text{s.t.} \quad & x \in \mathcal{X} \\ & \mathcal{R}_{\mathbb{P}}\big[f_i(x,\tilde{\xi})\big] \le 0 \qquad \forall i \in [I] \end{aligned}} \qquad \text{(SP)}$$

$\mathcal{X}$: the constraints that do *not* involve ξ, the same in both programs; f_i : the constraints that do uncertainty: a random vector $\tilde{\xi} \in \mathbb{R}^k$ with known distribution $\mathbb{P}$, $\tilde{\xi} \sim \mathbb{P}$ (ξ = one realization)

e.g. linear program: $\xi = (c, A, b)$, $\tilde{\xi} = (\tilde{c}, \tilde{A}, \tilde{b})$, $a_i^\top$ = row i of A, $\mathcal{X} = \{\, x : x \ge 0 \,\}$

$$\begin{aligned} \min \quad & c^\top x \\ \text{s.t.} \quad & x \in \mathcal{X} \\ & a_i^\top x - b_i \le 0 \quad \forall i \in [I] \end{aligned} \quad \longrightarrow \quad \begin{aligned} \min \quad & \mathcal{R}_{\mathbb{P}}\big[\tilde{c}^\top x\big] \\ \text{s.t.} \quad & x \in \mathcal{X} \\ & \mathcal{R}_{\mathbb{P}}\big[\tilde{a}_i^\top x - \tilde{b}_i\big] \le 0 \quad \forall i \in [I] \end{aligned}$$

fix x: $f_0(x,\xi)$ is a number; $f_0(x,\tilde{\xi})$ is a *random variable* — we cannot minimize it or ask it to be ≤ 0

In SP, each uncertain function is wrapped (i.e., evaluated) in a risk measure, which turns it into one number.

Def (risk measure) $\mathcal{R}_{\mathbb{P}} : \mathcal{L} \longrightarrow \mathbb{R}$ $\mathcal{L}$ = all univariate random variables, $\mathbb{R}$ = real numbers

the index $\mathbb{P}$ is the distribution of $\tilde{\xi}$: every risk measure depends on it

2. Uncertainty in the objective

Uncertainty only in the objective; the constraints are just $x \in \mathcal{X}$:

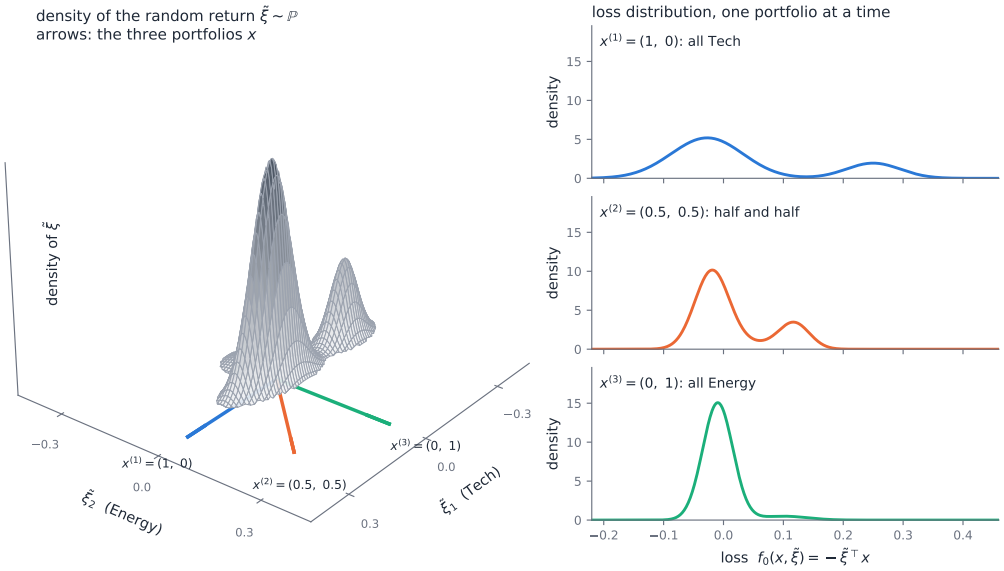
$$\min_{x \in \mathcal{X}} \ \mathcal{R}_{\mathbb{P}}\big[f_0(x, \tilde{\xi})\big]$$

$\tilde{Z} = f_0(x, \tilde{\xi})$ the loss, a univariate random variable

fix x $\Rightarrow$ $\tilde{Z}$ has a distribution; change x $\Rightarrow$ it changes — **the decision shapes the distribution**

Example (Portfolio optimization) $\tilde{\xi}$ = vector of asset returns

$$\begin{aligned} \min \quad & \mathcal{R}_{\mathbb{P}}\big[\underbrace{-\tilde{\xi}^\top x}_{= f_0(x, \tilde{\xi})}\big] \\ \text{s.t.} \quad & x \in \mathcal{X} = \{\, x \in \mathbb{R}^n \ : \ e^\top x = 1, \ x \geq 0 \,\} \end{aligned}$$



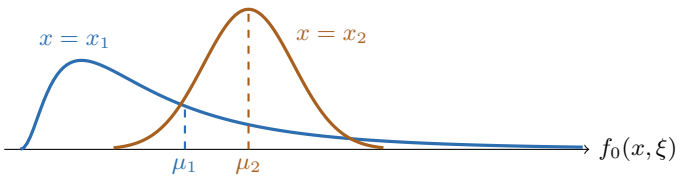
how do we compare the three? we need one number per distribution: the risk measure $\mathcal{R}_{\mathbb{P}}$.

2.1 Risk neutral

$\mathcal{R}_{\mathbb{P}} = \mathbb{E}_{\mathbb{P}}$: expected value

minimizing $\mathbb{E}_{\mathbb{P}}$ over x:

$$\min_{x \in \mathcal{X}} \ \mathbb{E}_{\mathbb{P}}\big[f_0(x, \tilde{\xi})\big]$$



$\mu_1 = \mathbb{E}_{\mathbb{P}}[f_0(x_1, \tilde{\xi})] \ < \ \mu_2 = \mathbb{E}_{\mathbb{P}}[f_0(x_2, \tilde{\xi})]$ $\Rightarrow$ x_1 is preferable under the risk-neutral measure.

2.2 Risk averse 1: mean–variance

$\mathcal{R}_{\mathbb{P}} = \lambda \operatorname{Var}_{\mathbb{P}} + (1-\lambda)\,\mathbb{E}_{\mathbb{P}}$: convex combination of mean and variance

minimizing mean–variance over x:

$$\min_{x\in\mathcal{X}} \quad \lambda \operatorname{Var}_{\mathbb{P}}\big[f_0(x,\tilde{\xi})\big] + (1-\lambda)\,\mathbb{E}_{\mathbb{P}}\big[f_0(x,\tilde{\xi})\big] \qquad \lambda \in [0,1]$$

Example (Mean–variance portfolio optimization) (Harry Markowitz, 1952)

$\tilde{\xi} \in \mathbb{R}^n$ is the vector of asset returns. Assume we know

$\mathbb{E}_{\mathbb{P}}[\tilde{\xi}] = \mu$ vector of means, $\mathbb{E}_{\mathbb{P}}\big[(\tilde{\xi}-\mu)(\tilde{\xi}-\mu)^\top\big] = \mathbb{E}_{\mathbb{P}}[\tilde{\xi}\tilde{\xi}^\top] - \mu\mu^\top = \Sigma$ covariance matrix

an investor allocates a wealth of \$1 to minimize the mean–variance of the portfolio loss $-\tilde{\xi}^\top x$:

$$\begin{array}{ll} \min & \lambda \operatorname{Var}_{\mathbb{P}}\big(-\tilde{\xi}^\top x\big) + (1-\lambda)\,\mathbb{E}_{\mathbb{P}}\big[-\tilde{\xi}^\top x\big] \\ \text{s.t.} & e^\top x = 1 \\ & x \geq 0 \end{array}$$

(Review: HW#2) **reformulate** — both terms are functions of μ and Σ only:

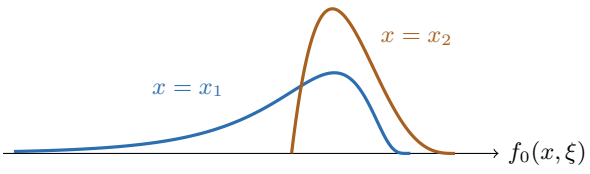
$\mathbb{E}_{\mathbb{P}}[-\tilde{\xi}^\top x] = -\mu^\top x$ linearity of expectation

$$\begin{aligned} \operatorname{Var}_{\mathbb{P}}(-\tilde{\xi}^\top x) = \operatorname{Var}_{\mathbb{P}}(\tilde{\xi}^\top x) &= \mathbb{E}_{\mathbb{P}}\big[(\tilde{\xi}^\top x)^2\big] - \big(\mathbb{E}_{\mathbb{P}}[\tilde{\xi}^\top x]\big)^2 \\ &= \mathbb{E}_{\mathbb{P}}\big[x^\top \tilde{\xi}\tilde{\xi}^\top x\big] - (\mu^\top x)^2 = x^\top \mathbb{E}_{\mathbb{P}}[\tilde{\xi}\tilde{\xi}^\top]\, x - x^\top \mu\mu^\top x \\ &= x^\top \big(\mathbb{E}_{\mathbb{P}}[\tilde{\xi}\tilde{\xi}^\top] - \mu\mu^\top\big) x \\ &= x^\top \Sigma\, x \end{aligned}$$

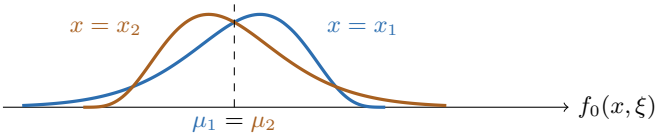
$$\begin{array}{lll} \min & \lambda\, x^\top \Sigma\, x - (1-\lambda)\,\mu^\top x & \\ \text{s.t.} & e^\top x = 1 & \Sigma \succeq 0 \;\Rightarrow\; \text{convex quadratic program} \\ & x \geq 0 & \end{array}$$

Limits of mean–variance

- for a general loss $f_0(x,\tilde{\xi})$, $\operatorname{Var}_{\mathbb{P}}[f_0(x,\tilde{\xi})]$ need not be convex in x — the mean–variance risk measure in general leads to a non-convex optimization problem. (the above model is an exception: $f_0(x,\xi) = -\xi^\top x$ is linear, hence the variance term leads to a quadratic function, hence convex)
- variance penalizes spread in *both* directions — it does not see where the big losses are:



x_2 has lower variance, but has higher probability of generating bigger losses.

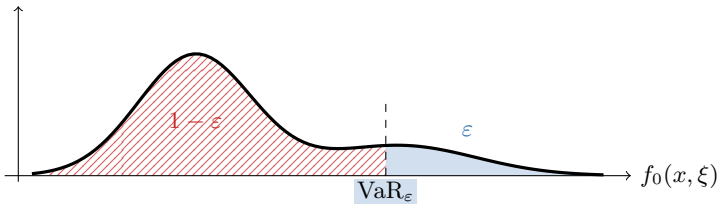


x_1 and x_2 have the same mean and the same variance (mirror images), so mean–variance cannot tell them apart; but x_1 is preferred because it has higher probability of generating smaller losses.

2.3 Risk averse 2: Value at Risk (VaR)

$\mathcal{R}_{\mathbb{P}} = \mathrm{VaR}_\varepsilon$: $(1-\varepsilon)$-quantile of the loss typically a non-convex optimization problem

Of course, VaR_ε depends on $\mathbb{P}$ just like $\mathbb{E}_{\mathbb{P}}$ and $\mathrm{Var}_{\mathbb{P}}$; we omit the index for notational simplicity.



— popular in finance

— imposed by the Basel II accord: how much capital banks need to set aside to hold against financial risks

Formally:

$$\mathrm{VaR}_\varepsilon\big[f_0(x,\tilde{\xi})\big] \;=\; \min_{t\in\mathbb{R}} \; t \quad \text{s.t.} \;\; \mathbb{P}\big(f_0(x,\tilde{\xi}) \le t\big) \ge 1-\varepsilon$$

the constraint is a chance / probability constraint: generally non-convex, highly intractable

the smallest loss level t that is exceeded with probability at most ε

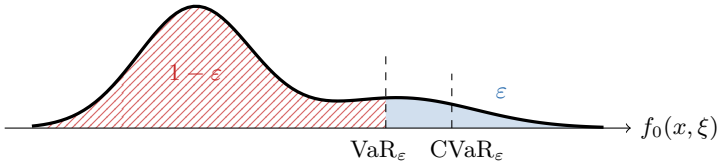
minimizing VaR over x: the min over t joins the outer min over x

$$\begin{aligned}\min_{x\in\mathcal{X}} \mathrm{VaR}_\varepsilon\big[f_0(x,\tilde{\xi})\big] &= \min_{x\in\mathcal{X}} \min_{t\in\mathbb{R}} \; t \quad \text{s.t.} \quad \mathbb{P}\big(f_0(x,\tilde{\xi}) \le t\big) \ge 1-\varepsilon \\ &= \min_{x\in\mathcal{X},\; t\in\mathbb{R}} \; t \quad \text{s.t.} \quad \mathbb{P}\big(f_0(x,\tilde{\xi}) \le t\big) \ge 1-\varepsilon\end{aligned}$$

VaR risk measure in general leads to a non-convex optimization problem.

2.4 Risk averse 3: Conditional Value at Risk (CVaR)

$\mathcal{R}_{\mathbb{P}} = \mathrm{CVaR}_\varepsilon$: average loss in the worst ε fraction of outcomes convex if f_0 is convex in x



with $\tilde{Z} = f_0(x,\tilde{\xi})$: CVaR is the average loss in the tail beyond VaR_ε, i.e., a conditional expectation:

Def (conditional expectation) $\quad \mathbb{E}_{\mathbb{P}}\big[\tilde{Z} \,\big|\, \tilde{Z} \in A\big] = \dfrac{\mathbb{E}_{\mathbb{P}}\big[\tilde{Z}\,\mathbb{I}[\tilde{Z}\in A]\big]}{\mathbb{P}(\tilde{Z}\in A)}$

Proposition: if $\tilde{Z}$ has a continuous distribution, then $\quad \mathrm{CVaR}_\varepsilon[\tilde{Z}] = \mathbb{E}_{\mathbb{P}}\big[\tilde{Z} \,\big|\, \tilde{Z} \ge \mathrm{VaR}_\varepsilon[\tilde{Z}]\big]$

here A is the tail, of probability ε: $\quad \mathrm{CVaR}_\varepsilon[\tilde{Z}] = \dfrac{1}{\varepsilon}\,\mathbb{E}_{\mathbb{P}}\big[\tilde{Z}\,\mathbb{I}[\tilde{Z} \ge \mathrm{VaR}_\varepsilon[\tilde{Z}]]\big]$

— CVaR = the **average loss in the worst ε fraction** of outcomes; so $\mathrm{CVaR}_\varepsilon \ge \mathrm{VaR}_\varepsilon$.

— introduced by Rockafellar & Uryasev (2000)

— imposed by the Basel III accord

Formally:

$$\boxed{\mathrm{CVaR}_\varepsilon\big[f_0(x,\tilde{\xi})\big] \;=\; \min_{\beta\in\mathbb{R}} \;\; \beta + \frac{1}{\varepsilon}\,\mathbb{E}_{\mathbb{P}}\Big[\max\big\{f_0(x,\tilde{\xi})-\beta,\;0\big\}\Big]}$$

this formula works for *any* distribution of $\tilde{Z}$; for a continuous $\tilde{Z}$ it equals the conditional expectation above

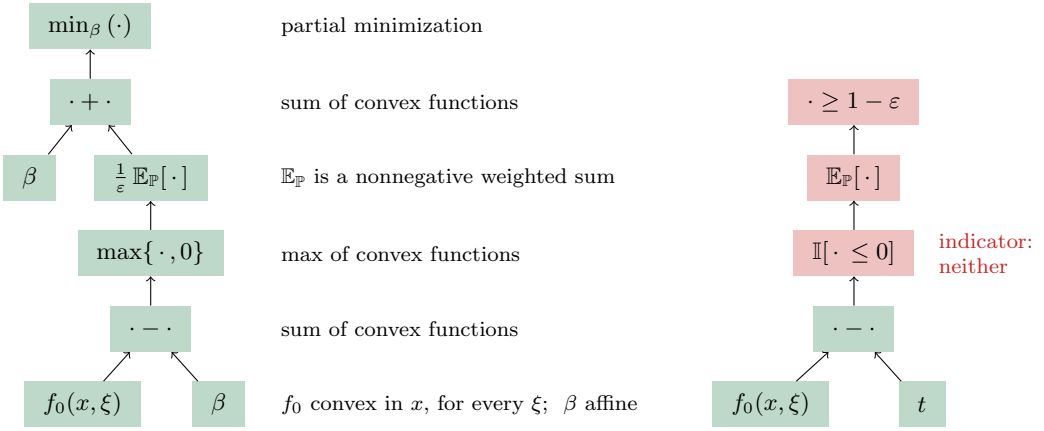
minimizing CVaR over x: the min over β joins the outer min over x

$$\begin{aligned}\min_{x\in\mathcal{X}} \mathrm{CVaR}_\varepsilon\big[f_0(x,\tilde{\xi})\big] &= \min_{x\in\mathcal{X}} \min_{\beta\in\mathbb{R}} \;\; \beta + \frac{1}{\varepsilon}\,\mathbb{E}_{\mathbb{P}}\Big[\max\big\{f_0(x,\tilde{\xi})-\beta,\;0\big\}\Big] \\ &= \min_{x\in\mathcal{X},\;\beta\in\mathbb{R}} \;\; \beta + \frac{1}{\varepsilon}\,\mathbb{E}_{\mathbb{P}}\Big[\max\big\{f_0(x,\tilde{\xi})-\beta,\;0\big\}\Big]\end{aligned}$$

the objective is convex in (x,β) when f_0 is convex in x: $\max\{\text{convex},0\}$ is convex, and $\mathbb{E}$ keeps it convex.

why CVaR is convex: build it from convex pieces, with the operations from HW1 and week 2

CVaR $= \min_\beta \;\beta + \frac{1}{\varepsilon}\,\mathbb{E}_{\mathbb{P}}\big[\max\{\tilde{Z}-\beta,\;0\}\big]$ **VaR** $= \min_t \;t$ s.t. $\mathbb{P}\big(\tilde{Z}\le t\big)\ge 1-\varepsilon$



green: convex or affine, jointly in (x,β) or (x,t) • red: neither

for VaR, the chance constraint leads to the indicator, $\mathbb{P}(\tilde{Z}\le t)=\mathbb{E}_{\mathbb{P}}\big[\mathbb{I}[\tilde{Z}\le t]\big]\ge 1-\varepsilon$, i.e., a MIP.
here $\mathbb{E}_{\mathbb{P}}\big[\mathbb{I}[A]\big]=1\cdot\mathbb{P}(A)+0\cdot\big(1-\mathbb{P}(A)\big)=\mathbb{P}(A)$, since $\mathbb{I}[A]$ is 1 exactly when A holds

3. Uncertainty in the constraint

Now uncertainty only in the constraints; the objective $f_0(x)$ is deterministic:

$$\min_{x\in\mathcal{X}} \; f_0(x) \quad \text{s.t.} \quad \mathcal{R}_{\mathbb{P}}\big[f_i(x,\tilde{\xi})\big] \le 0 \quad \forall i \in [I]$$

Three choices of $\mathcal{R}_{\mathbb{P}}$:

— $\mathbb{E}_{\mathbb{P}}\big[f_i(x,\tilde{\xi})\big] \le 0$ the constraint holds *on average*

— $\mathrm{VaR}_{\varepsilon}\big[f_i(x,\tilde{\xi})\big] \le 0$ non-convex in general

$\iff \min_{t\in\mathbb{R}} \big\{ t \;:\; \mathbb{P}\big(f_i(x,\tilde{\xi}) \le t\big) \ge 1-\varepsilon \big\} \le 0$

$\iff \exists\, t \le 0 \;:\; \mathbb{P}\big(f_i(x,\tilde{\xi}) \le t\big) \ge 1-\varepsilon$

$\iff \mathbb{P}\big(f_i(x,\tilde{\xi}) \le 0\big) \ge 1-\varepsilon$

Chance constraint / probabilistic constraint $\mathbb{P}\big(f_i(x,\tilde{\xi}) \le 0\big) \ge 1-\varepsilon$

Proposition: $\mathrm{VaR}_{\varepsilon}$ constraint $\iff$ chance constraint

the chance constraint from week 10 is a VaR constraint: "violated with probability at most ε" is the same as "the $(1-\varepsilon)$-quantile of f_i is at most 0".

the full problem:

$$\begin{array}{ll} \min\limits_{x\in\mathcal{X}} & f_0(x) \\ \text{s.t.} & \mathrm{VaR}_{\varepsilon}\big[f_i(x,\tilde{\xi})\big] \le 0 \quad \forall i \in [I] \end{array} \iff \begin{array}{ll} \min\limits_{x\in\mathcal{X}} & f_0(x) \\ \text{s.t.} & \mathbb{P}\big(f_i(x,\tilde{\xi}) \le 0\big) \ge 1-\varepsilon \quad \forall i \in [I] \end{array}$$

— $\mathrm{CVaR}_{\varepsilon}\big[f_i(x,\tilde{\xi})\big] \le 0$ convex; gives an upper bound on $\mathrm{VaR}_{\varepsilon}$, so it implies the VaR constraint

$\iff \min_{\beta}\; \beta + \dfrac{1}{\varepsilon}\,\mathbb{E}_{\mathbb{P}}\big[\max\{f_i(x,\tilde{\xi}) - \beta, 0\}\big] \le 0$

$\iff \exists\, \beta \in \mathbb{R} \;:\; \beta + \dfrac{1}{\varepsilon}\,\mathbb{E}_{\mathbb{P}}\big[\max\{f_i(x,\tilde{\xi}) - \beta, 0\}\big] \le 0$

the full problem — the $\exists\,\beta$ becomes a decision variable, one β_i per constraint

$$\begin{array}{ll} \min\limits_{x\in\mathcal{X}} & f_0(x) \\ \text{s.t.} & \mathrm{CVaR}_{\varepsilon}\big[f_i(x,\tilde{\xi})\big] \le 0 \;\; \forall i \in [I] \end{array} \iff \begin{array}{ll} \min\limits_{x\in\mathcal{X},\; \beta\in\mathbb{R}^I} & f_0(x) \\ \text{s.t.} & \beta_i + \dfrac{1}{\varepsilon}\,\mathbb{E}_{\mathbb{P}}\big[\max\{f_i(x,\tilde{\xi}) - \beta_i, 0\}\big] \le 0 \;\; \forall i \in [I] \end{array}$$

CVaR is a tractable conservative approximation of VaR, hence of the chance constraint

for every x: $\mathrm{VaR}_{\varepsilon}[f_i] \le \mathrm{CVaR}_{\varepsilon}[f_i]$, since the average loss in the tail is at least where the tail starts

so imposing $\mathrm{CVaR}_{\varepsilon}[f_i] \le 0$ forces $\mathrm{VaR}_{\varepsilon}[f_i] \le 0$, which is the chance constraint $\mathbb{P}(f_i \le 0) \ge 1-\varepsilon$

— **tractable:** CVaR is convex when f_i is convex in x; the chance constraint is NP-hard in general

— **conservative:** every x feasible for CVaR is feasible for the chance constraint, but not vice versa

4. Uncertainty in both: the CVaR stochastic program

one β per risk term: β_0 for the objective, β_i for constraint i; all of them join the outer min

$$\begin{aligned} \min_{x\in\mathcal{X},\ \beta_0,\dots,\beta_I\in\mathbb{R}} \quad & \beta_0 + \frac{1}{\varepsilon}\,\mathbb{E}_{\mathbb{P}}\Big[\max\big\{f_0(x,\tilde{\xi}) - \beta_0,\ 0\big\}\Big] \\ \text{s.t.} \quad & \beta_i + \frac{1}{\varepsilon}\,\mathbb{E}_{\mathbb{P}}\Big[\max\big\{f_i(x,\tilde{\xi}) - \beta_i,\ 0\big\}\Big] \le 0 \qquad \forall i \in [I] \end{aligned}$$

convex whenever every f_i is convex in x.

with the discrete $\mathbb{P}$ from the recap, replace each max by a variable $u_{is} \ge 0$

$$\begin{aligned} \min \quad & \beta_0 + \frac{1}{\varepsilon}\sum_{s\in[S]} p_s\, u_{0s} \\ \text{s.t.} \quad & \beta_i + \frac{1}{\varepsilon}\sum_{s\in[S]} p_s\, u_{is} \le 0 \qquad\qquad \forall i \in [I] \\ & u_{is} \ge f_i(x,\xi^s) - \beta_i, \quad u_{is} \ge 0 \quad \forall i \in \{0\}\cup[I],\ s \in [S] \\ & x \in \mathcal{X} \end{aligned}$$

linear f_i and polyhedral $\mathcal{X}$ $\Rightarrow$ a linear program, with no binary variables. The VaR / chance constraint needs a binary variable per scenario (which scenarios may violate the constraint) — the MIP in the recap. That is why CVaR is the tractable one.